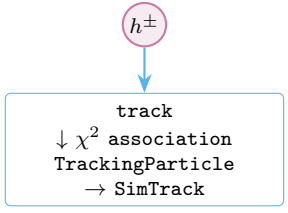
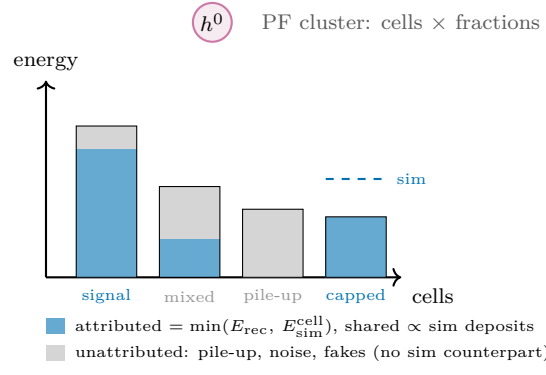
charged candidate $h^{\pm}$



match $\Rightarrow$ **100%** of the candidate energy is signal
no match $\Rightarrow$ **0%** (pile-up track, fake)

TrackingParticles exist for the hard-scatter event only

neutral candidate γ / h^0



sum over the cells, rescaled to the candidate energy
$\Rightarrow$ `signalFraction` of the candidate
the sim hits exist for the hard-scatter event only

the jet

$$\vec{p}_T^{\,\text{sig}} = \sum_{c \,\in\, \text{jet}} w_c \, f_c^{\text{sig}} \, \vec{p}_{T,c}$$

w_c: clustering weight (Puppi), f_c^{sig}: `signalFraction`

`signalPtRatio` $= |\vec{p}_T^{\,\text{sig}}|/p_T^{\text{GEN}}$

`ptRatio` $-$ `signalPtRatio` $=$ the pile-up in this jet

shared-energy response: keep only the shares whose particle descends from the matched GenJet's constituents
$\Rightarrow$ the gen jet's own energy, nothing else

signal = any particle of the hard-scatter event (underlying event and other jets included); nothing is subtracted, pile-up is simply never counted